

Workshop 6

Problem 1

Keywords: Displacement, Velocity, Acceleration.

The position $f(t)$ of an object moving vertically is given by

$$f(t) = \frac{1}{4} \cos t + 4 \sin t - 4t \cos t,$$

with t in seconds and $f(t)$ in feet. We assume that the movement starts at $t = 0$. In this problem, the vertical axis is oriented downward.

- (a) Find the velocity $v(t)$ of the object.
- (b) Find the acceleration $a(t)$ of the object.
- (c) Show that the position $f(t)$ satisfies the equation

$$f''(t) + f(t) = 8 \sin(t).$$

- (d) Find the first time interval the object is moving upward. Find the first time interval the object is moving downward.
- (e) Find the displacement of the object on the first time interval it is moving upward.

Problem 2

Keywords: Chain Rule, Find the Error

Students were asked to compute the derivative of $f(x) = \sin(e^x - x^2 + 2)$.

- (a) Find the derivative.
- (b) Each of the following submissions for $f'(x)$ is incorrect. Write a sentence about what each student did wrong on the problem and what they should remember or think about in order to get a similar problem correct in the future.

Student 1: $f'(x) = \cos(e^x - x^2 + 2)$

Student 2: $f'(x) = \cos(e^x - 2x)$

Student 3: $f'(x) = \cos x(e^x - 2x)$

Student 4: $f'(x) = \cos(e^x - x^2 + 2)(e^x - x^2 + 2)$

Problem 3

Keywords: Chain Rule, Equations of Tangent Lines.

Let f be a differentiable function on $(-\infty, +\infty)$. Assume that we know the following.

- $f(2) = 9$, $f'(2) = 6$, $f'(1) = 4$.

- The tangent line to $y = f(x^2)$ at $x = -3$ has equation $y = -12x - 5$.

Determine the following if possible. If there is not enough information to answer the question, explain why in 1-2 English sentences, specifying what information is missing.

- (a) An equation of the tangent line to f at $x = 9$.
- (b) An equation of the tangent line to $y = f(f(x))$ at $x = 2$.
- (c) The value of $\frac{d}{dx} (xf(2x^2 - x))|_{x=1}$.
- (d) The value of $\frac{d}{dx} (\cos(\pi x)f(2x^2 - x))|_{x=1}$.
- (e) An equation of the tangent line to $y = \cos(\pi x)f(2x^2 - x)$ at $x = 1$.